weygandt
kimmel
kieso
team for success



ACCOUNTING PRINCIPLES

TWELFTH EDITION

Coby Harmon
University of California, Santa Barbara
Westmont College

WILEY

5 Accounting for Merchandising Operations

Learning Objectives

1. Describe merchandising operations and inventory systems.
2. Record purchases under a perpetual inventory system.
3. Record sales under a perpetual inventory system.
4. Apply the steps in the accounting cycle to a merchandising company.
5. Compare a multiple-step with a single-step income statement.

LEARNING OBJECTIVE 1

Describe merchandising operations and inventory systems.

Merchandising Companies

Buy and Sell Goods

Retailer



Wholesaler





Consumer

The primary source of revenues is referred to as **sales revenue** or **sales**.

Merchandising Operations

Income Measurement

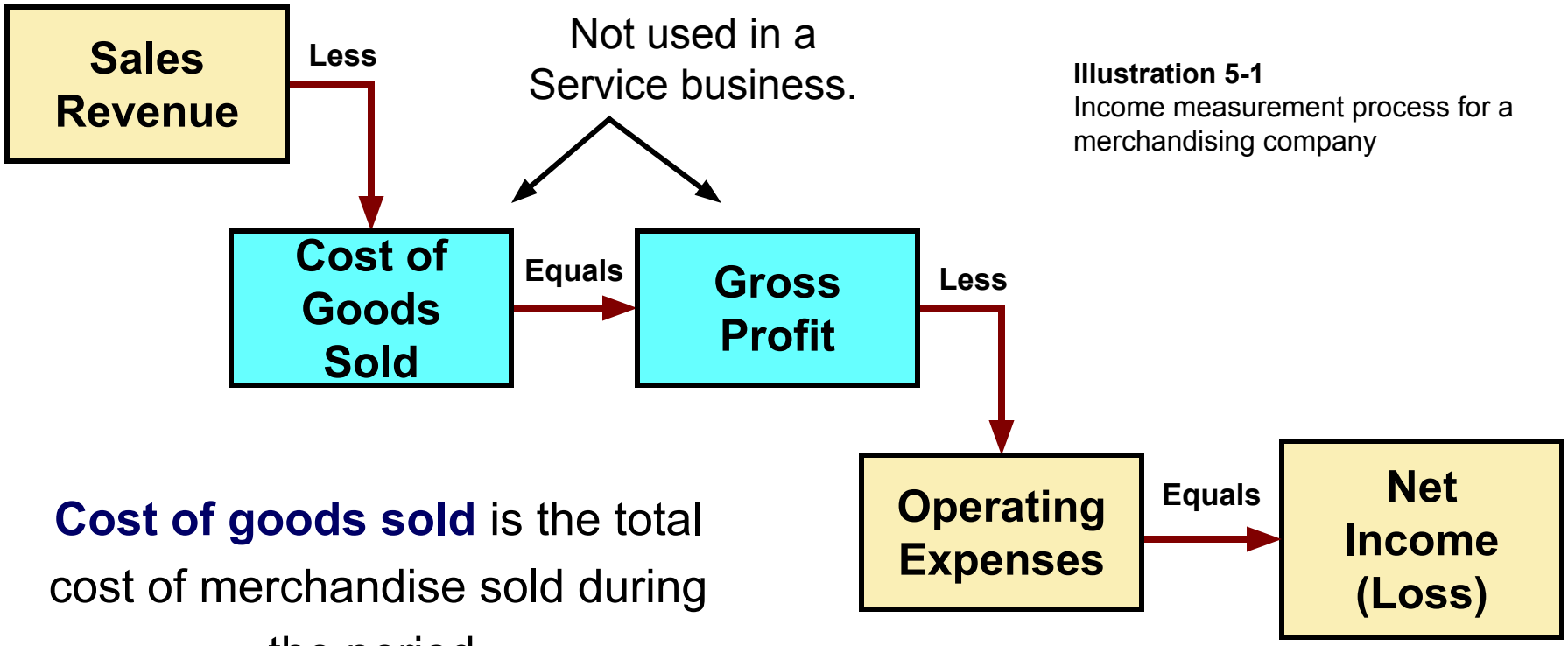


Illustration 5-1
Income measurement process for a merchandising company

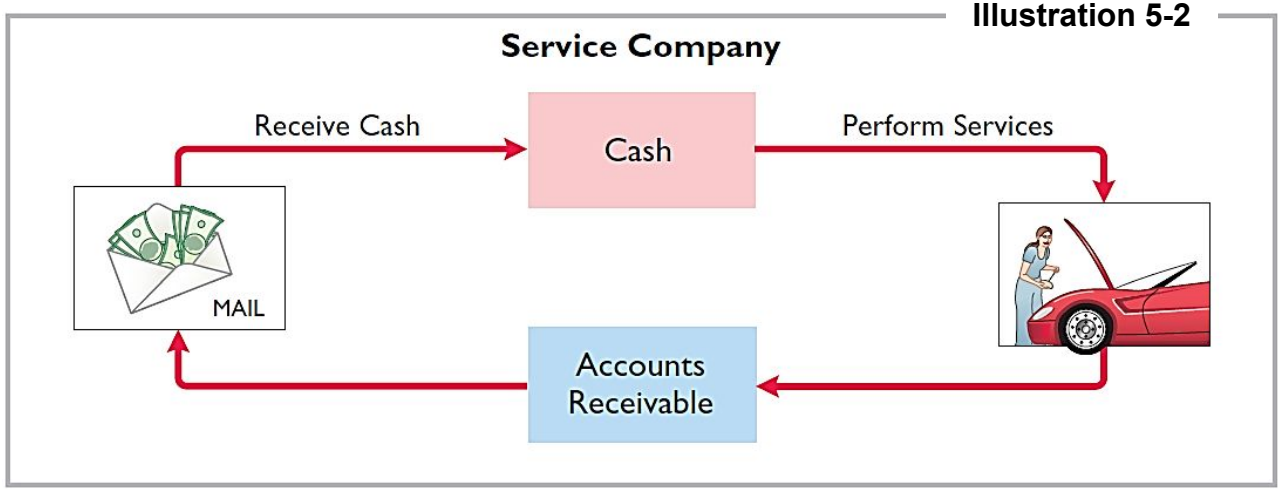
Cost of goods sold is the total cost of merchandise sold during the period.

LO 1

Operating Cycles

The operating cycle of a **merchandising company** ordinarily is longer than that of a **service company**.

Illustration 5-2



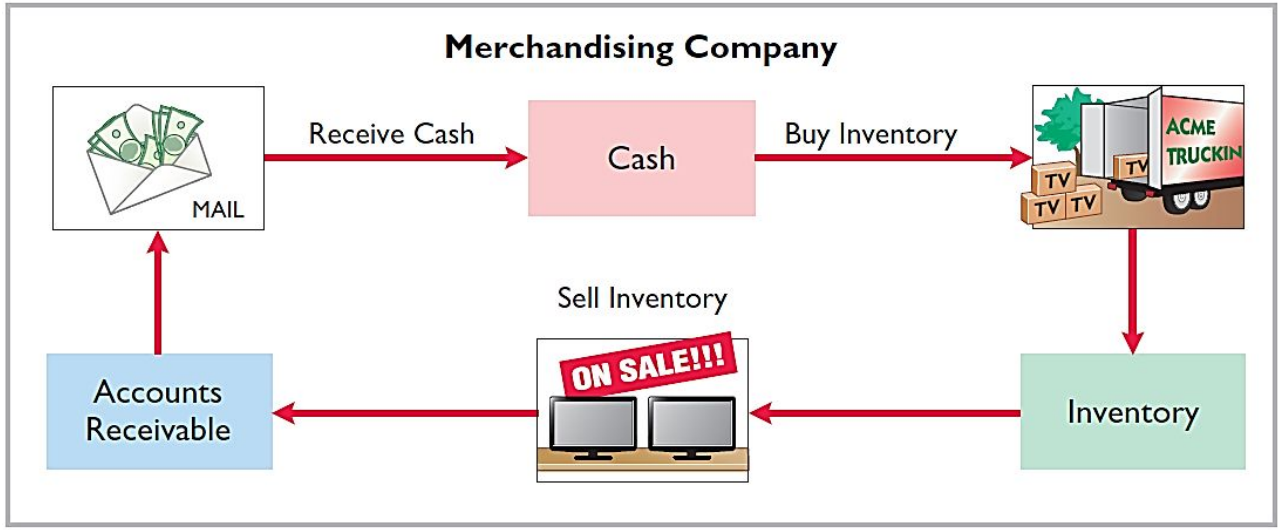
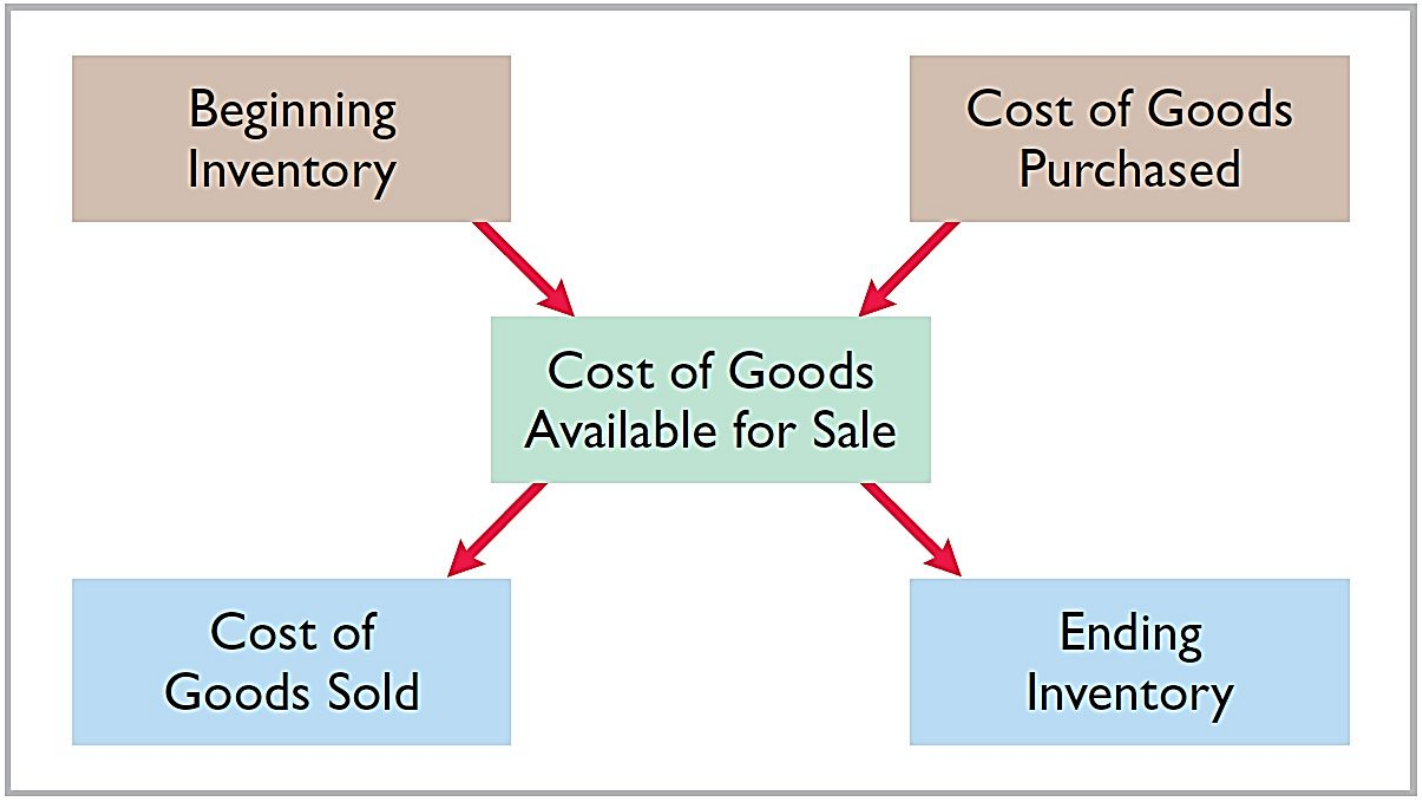


Illustration 5-3

LO 1

Flow of Costs

Illustration 5-4



Companies use either a **perpetual inventory system** or a **periodic inventory system** to account for inventory.

LO 1

Flow of Costs

PERPETUAL SYSTEM

- Maintain detailed records of the cost of each inventory purchase and sale.
- Records continuously show inventory that should be on hand for every item.
- Company determines cost of goods sold each time a sale occurs.

Flow of Costs

PERIODIC SYSTEM

- Do not keep detailed records of the goods on hand.
- Cost of goods sold determined by count at the end of the accounting period.
- Calculation of Cost of Goods Sold:

Beginning inventory	$ 100,000
Add: Purchases, net	800,000
Goods available for sale	900,000
Less: Ending inventory	125,000
Cost of goods sold	$ 775,000

Flow of Costs

ADVANTAGES OF THE PERPETUAL SYSTEM

- Traditionally used for merchandise with high unit values.
- Shows the quantity and cost of the inventory that should be on hand at any time.
- Provides better control over inventories than a periodic system.

LO 1

LEARNING OBJECTIVE 2

Record purchases under a perpetual inventory system.

- Made using **cash or credit** (on account).
- Normally **record when** goods are received from the seller.
- **Purchase invoice** should support each credit purchase.

INVOICE NO. 731

PW AUDIO SUPPLY, INC.
27 CIRCLE DRIVE
HARDING, MICHIGAN 48281

SOLD TO
Firm Name: Sauk Stereo
Attention of: James Hoover, Purchasing Agent
Address: 125 Main Street
Chelsea (City) Illinois (State) 60915 (Zip)

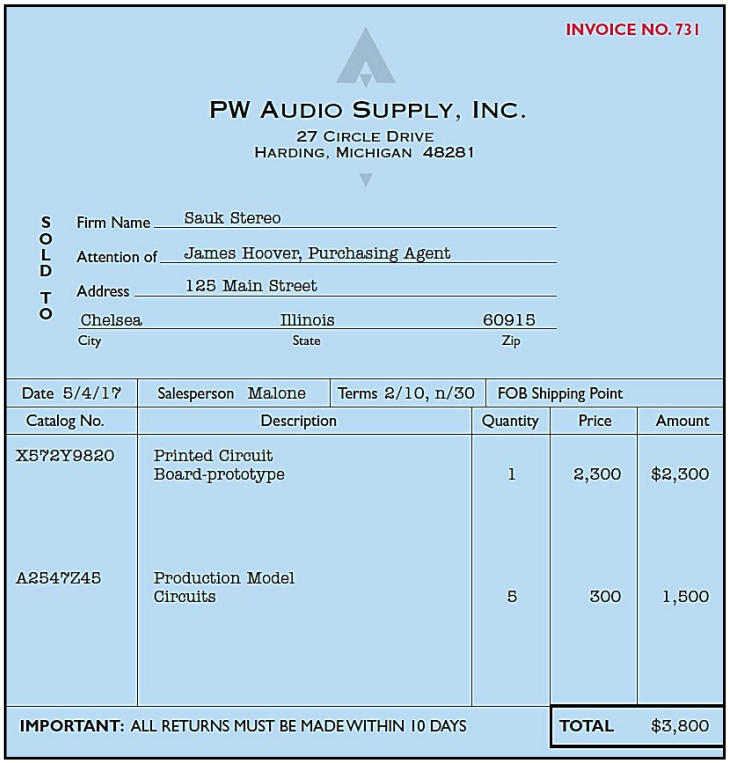
Date 5/4/17	Salesperson Malone	Terms 2/10, n/30	FOB Shipping Point	
Catalog No.	Description	Quantity	Price	Amount
X572Y9820	Printed Circuit Board-prototype	1	2,300	$2,300
A2547Z45	Production Model Circuits	5	300	1,500
IMPORTANT: ALL RETURNS MUST BE MADE WITHIN 10 DAYS			TOTAL	$3,800

Illustration 5-6
Sales invoice used as purchase invoice by Sauk Stereo

Recording Purchases of Merchandise

Illustration: Sauk Stereo (the buyer) uses as a purchase invoice the sales invoice prepared by PW Audio Supply, Inc. (the seller). **Prepare the journal entry** for Sauk Stereo for the invoice from PW Audio Supply.

Illustration 5-6



INVOICE NO. 731

PW AUDIO SUPPLY, INC.
27 CIRCLE DRIVE
HARDING, MICHIGAN 48281

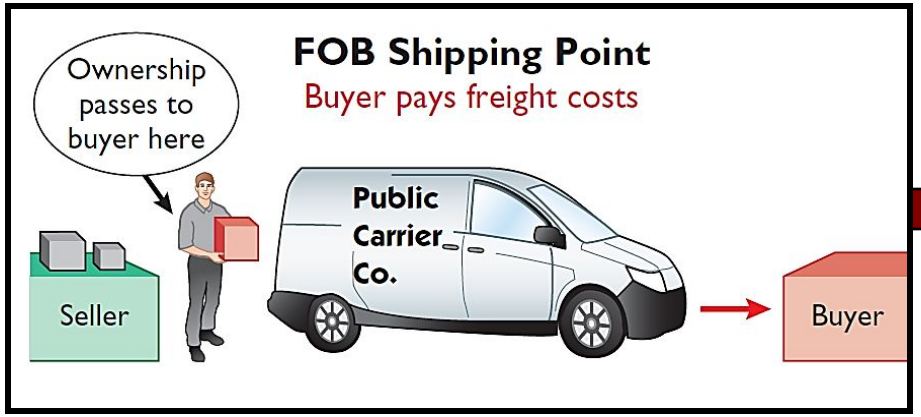
SOLD TO
Firm Name: Sauk Stereo
Attention of: James Hoover, Purchasing Agent
Address: 125 Main Street
Chelsea (City) Illinois (State) 60915 (Zip)

Date 5/4/17	Salesperson Malone	Terms 2/10, n/30	FOB Shipping Point	
Catalog No.	Description	Quantity	Price	Amount
X572Y9820	Printed Circuit Board-prototype	1	2,300	$2,300
A2547Z45	Production Model Circuits	5	300	1,500
IMPORTANT: ALL RETURNS MUST BE MADE WITHIN 10 DAYS			TOTAL	$3,800

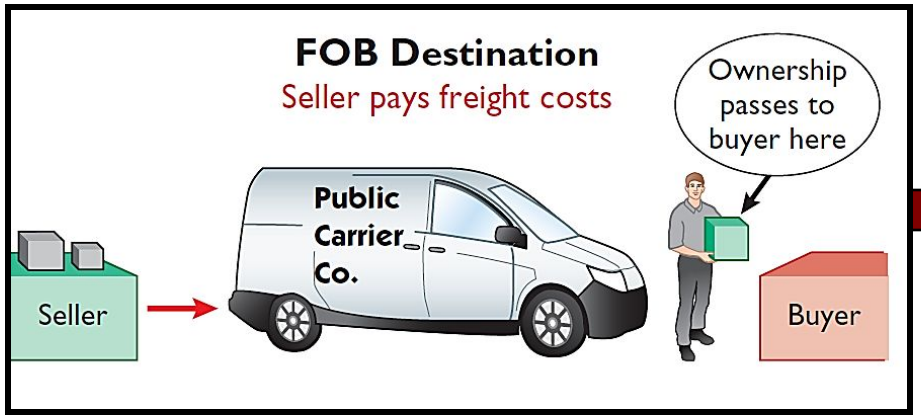
May 4	Inventory	3,800	
	Accounts Payable		3,800

LO 2

Freight Costs



Ownership of the goods passes to the buyer when the public carrier accepts the goods from the seller.



Ownership of the goods remains with the seller until the goods reach the buyer.

Illustration 5-7
Shipping terms

Freight costs incurred by the seller are an **operating expense**.

LO 2

Freight Costs

Illustration: Assume upon delivery of the goods on May 6, **Sauk Stereo pays** Public Freight Company $150 for **freight charges**, the entry on Sauk Stereo's books is:

May 6	Inventory	150	
	Cash		150

Assume the freight terms on the invoice in Illustration 5-6 had required **PW Audio Supply to pay the freight charges**, the entry by PW Audio Supply would have been:

May 4	Freight-Out	150	
	Cash		150

LO 2

Purchase Returns and Allowances

Purchaser may be dissatisfied because goods are damaged or defective, of inferior quality, or do not meet specifications.

Purchase Return	Purchase Allowance
Return goods for credit if the sale was made on credit, or for a cash refund if the purchase was for cash.	May choose to keep the merchandise if the seller will grant a reduction of the purchase price.

Purchase Returns and Allowances

Illustration: Assume Sauk Stereo returned goods costing $300 to PW Audio Supply on May 8.

May 8	Accounts Payable	300	
	Inventory		300

Purchase Discounts

Credit terms may permit buyer to claim a cash discount for prompt payment.

Example: Credit terms may read **2/10, n/30**.

Advantages:

- Purchaser saves money.
- Seller shortens the operating cycle by converting the accounts receivable into cash earlier.

Purchase Discounts

2/10, n/30	1/10 EOM	n/10 EOM
2% discount if paid within 10 days, otherwise net amount due within 30 days.	1% discount if paid within first 10 days of next month.	Net amount due within the first 10 days of the next month.

Purchase Discounts

Illustration: Assume Sauk Stereo pays the balance due of $3,500 (gross invoice price of $3,800 less purchase returns and allowances of $300) on May 14, the last day of the discount period. Prepare the journal entry Sauk Stereo makes on May 14 to record the payment.

May 14	Accounts Payable	3,500	
	Inventory		70
	Cash		3,430

(Discount = $3,500 x 2% = **$70**)

Purchase Discounts

Illustration: If Sauk Stereo failed to take the discount, and instead made full payment of $3,500 on June 3, the journal entry would be:

June 3	Accounts Payable	3,500	
	Cash		3,500

Purchase Discounts

Should discounts be taken when offered?

Discount of 2% on $3,500	$	70.00
$3,500 invested at 10% for 20 days		19.18
Savings by taking the discount	**$**	**50.82**

Example: 2% for 20 days = Annual rate of 36.5%

$3,500 x 36.5% x 20 ÷ 365 = $70

Summary of Purchasing Transactions

Inventory

	Debit	Credit	
4th - Purchase	3,800	300	8th - Return
6th – Freight-in	150	70	14th - Discount
Balance	3,580		

LO 2

LEARNING OBJECTIVE 3

Record sales under a perpetual inventory system.

- Made using **cash or credit** (on account).
- Sales revenue, like service revenue, is recorded when the performance obligation is satisfied.
- Performance obligation is satisfied when the goods are transferred from the seller to the buyer.
- Sales invoice should support each credit sale.

Illustration 5-6

INVOICE NO. 731

PW AUDIO SUPPLY, INC.
27 CIRCLE DRIVE
HARDING, MICHIGAN 48281

SOLD TO
Firm Name: Sauk Stereo
Attention of: James Hoover, Purchasing Agent
Address: 125 Main Street
Chelsea (City) Illinois (State) 60915 (Zip)

Date 5/4/17	Salesperson Malone	Terms 2/10, n/30	FOB Shipping Point	
Catalog No.	Description	Quantity	Price	Amount
X572Y9820	Printed Circuit Board-prototype	1	2,300	$2,300
A2547Z45	Production Model Circuits	5	300	1,500
IMPORTANT: ALL RETURNS MUST BE MADE WITHIN 10 DAYS			TOTAL	$3,800

Recording Sales of Merchandise

Journal Entries to Record a Sale

#1

Cash **or** Accounts receivable	XXX	
Sales revenue		XXX

Selling Price

#2

Cost of goods sold	XXX	
Inventory		XXX

Cost

Recording Sales of Merchandise

Illustration: PW Audio Supply records the sale of $3,800 on May 4 to Sauk Stereo on account (Illustration 5-6) as follows (assume the merchandise cost PW Audio Supply $2,400).

May 4	Accounts Receivable	3,800	
	Sales Revenue		3,800
4	Cost of Goods Sold	2,400	
	Inventory		2,400

Sales Returns and Allowances

- **"Flip side"** of purchase returns and allowances.
- **Contra-revenue account** to Sales Revenue (debit).
- **Sales not reduced** (debited) because:
  - Would obscure importance of sales returns and allowances as a percentage of sales.
  - Could distort comparisons.

Sales Returns and Allowances

Illustration: Prepare the entry PW Audio Supply would make to record the credit for returned goods that had a $300 selling price (assume a $140 cost). Assume the **goods were not defective**.

May 8	Sales Returns and Allowances	300	
	Accounts Receivable		300
8	Inventory	140	
	Cost of Goods Sold		140

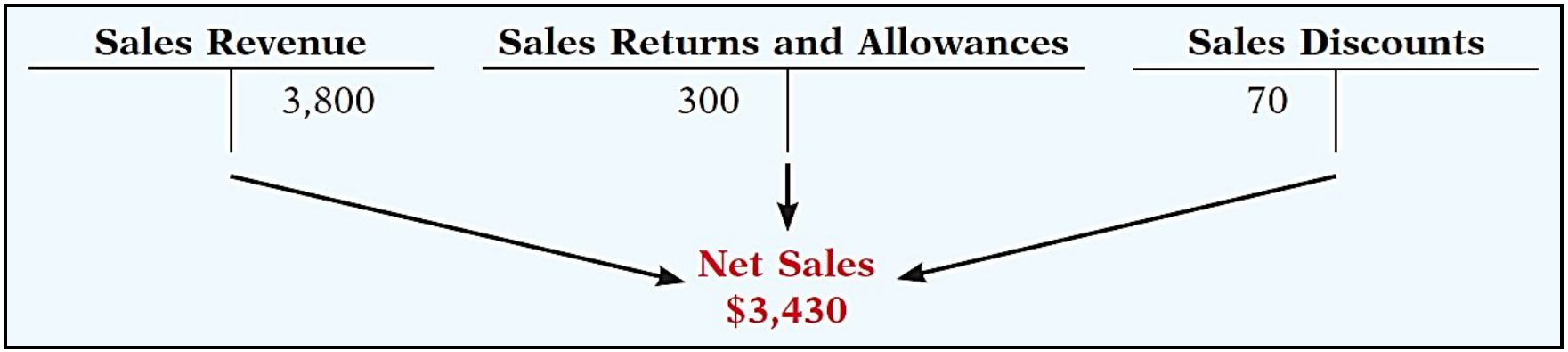
Sales Returns and Allowances

Illustration: Assume the **returned goods were defective** and had a scrap value of $50, PW Audio would make the following entries:

May 8	Sales Returns and Allowances	300	
	Accounts Receivable		300
8	Inventory	50	
	Cost of Goods Sold		50

Sales Discount

- Offered to customers to **promote prompt payment** of the balance due.
- **Contra-revenue account** (debit) to Sales Revenue.



LO 3

Sales Discount

Illustration: Assume Sauk Stereo pays the balance due of $3,500 (gross invoice price of $3,800 less purchase returns and allowances of $300) on May 14, the last day of the discount period. Prepare the journal entry PW Audio Supply makes to record the receipt on May 14.

May 14	Cash	3,430		
	Sales Discounts	70		*
	Accounts Receivable		3,500	

* [($3,800 – $300) X 2%]

LEARNING OBJECTIVE 4

Apply the steps in the accounting cycle to a merchandising company.

Adjusting Entries

- Generally the same as a service company.
- One additional adjustment to make the records agree with the actual inventory on hand.
- Involves adjusting Inventory and Cost of Goods Sold.

Adjusting Entries

Illustration: Suppose that PW Audio Supply has an unadjusted balance of $40,500 in Merchandise Inventory. Through a physical count, PW Audio determines that its actual merchandise inventory at year-end is $40,000. The company would make an adjusting entry as follows.

Cost of Goods Sold	500	
Inventory		500

Closing Entries

Dec. 31	Sales Revenue	480,000	
	Income Summary		480,000
	(To close income statement accounts with credit balances)		
31	Income Summary	450,000	
	Sales Returns and Allowances		12,000
	Sales Discounts		8,000
	Cost of Goods Sold		316,000
	Salaries and Wages Expense		64,000
	Freight-Out		7,000
	Advertising Expense		16,000
	Utilities Expense		17,000
	Depreciation Expense		8,000
	Insurance Expense		2,000
	(To close income statement accounts with debit balances)		

Closing Entries

31	Income Summary	30,000	
	Owner's Capital		30,000
	(To transfer net income to capital)		
31	Owner's Capital	15,000	
	Owner's Drawings		15,000
	(To close drawings to capital)		

LEARNING OBJECTIVE 5

Compare a multiple-step with a single-step income statement.

Multiple-Step Income Statement

- Shows several steps in determining net income.
- Two steps relate to principal operating activities.
- Distinguishes between operating and non-operating activities.

Multiple-Step

Key Items:

- **Net sales**

PW AUDIO SUPPLY
Income Statement
For the Year Ended December 31, 2017

Sales		
Sales revenue		$480,000
Less: Sales returns and allowances	$12,000	
Sales discounts	8,000	20,000
Net sales		460,000
Cost of goods sold		316,000
Gross profit		144,000
Operating expenses		
Salaries and wages expense	64,000	
Utilities expense	17,000	
Advertising expense	16,000	
Depreciation expense	8,000	
Freight-out	7,000	
Insurance expense	2,000	
Total operating expenses		114,000
Income from operations		30,000
Other revenues and gains		
Interest revenue	3,000	
Gain on disposal of plant assets	600	3,600
Other expenses and losses		
Interest expense	1,800	
Casualty loss from vandalism	200	2,000
Net income		$ 31,600

Illustration 5-14

LO 5

Multiple-Step

Key Items:

- Net sales
- **Gross profit**

PW AUDIO SUPPLY
Income Statement
For the Year Ended December 31, 2017

Sales		
Sales revenue		$480,000
Less: Sales returns and allowances	$12,000	
Sales discounts	8,000	20,000
Net sales		460,000
Cost of goods sold		316,000
Gross profit		144,000
Operating expenses		
Salaries and wages expense	64,000	
Utilities expense	17,000	
Advertising expense	16,000	
Depreciation expense	8,000	
Freight-out	7,000	
Insurance expense	2,000	
Total operating expenses		114,000
Income from operations		30,000
Other revenues and gains		
Interest revenue	3,000	
Gain on disposal of plant assets	600	3,600
Other expenses and losses		
Interest expense	1,800	
Casualty loss from vandalism	200	2,000
Net income		$ 31,600

Illustration 5-14

LO 5

Multiple-Step

Key Items:

- Net sales
- Gross profit
- **Operating expenses**

PW AUDIO SUPPLY
Income Statement
For the Year Ended December 31, 2017

Sales		
Sales revenue		$480,000
Less: Sales returns and allowances	$12,000	
Sales discounts	8,000	20,000
Net sales		460,000
Cost of goods sold		316,000
Gross profit		144,000
Operating expenses		
Salaries and wages expense	64,000	
Utilities expense	17,000	
Advertising expense	16,000	
Depreciation expense	8,000	
Freight-out	7,000	
Insurance expense	2,000	
Total operating expenses		114,000
Income from operations		30,000
Other revenues and gains		
Interest revenue	3,000	
Gain on disposal of plant assets	600	3,600
Other expenses and losses		
Interest expense	1,800	
Casualty loss from vandalism	200	2,000
Net income		$ 31,600

Illustration 5-14

LO 5

Multiple-Step

Key Items:

- Net sales
- Gross profit
- Operating expenses
- **Nonoperating activities**

PW AUDIO SUPPLY
Income Statement
For the Year Ended December 31, 2017

Sales		
Sales revenue		$480,000
Less: Sales returns and allowances	$12,000	
Sales discounts	8,000	20,000
Net sales		460,000
Cost of goods sold		316,000
Gross profit		144,000
Operating expenses		
Salaries and wages expense	64,000	
Utilities expense	17,000	
Advertising expense	16,000	
Depreciation expense	8,000	
Freight-out	7,000	
Insurance expense	2,000	
Total operating expenses		114,000
Income from operations		30,000
Other revenues and gains		
Interest revenue	3,000	
Gain on disposal of plant assets	600	3,600
Other expenses and losses		
Interest expense	1,800	
Casualty loss from vandalism	200	2,000
Net income		$ 31,600

Illustration 5-14

LO 5

Multiple-Step

Key Items:

- Net sales
- Gross profit
- Operating expenses
- **Nonoperating activities**

PW AUDIO SUPPLY
Income Statement
For the Year Ended December 31, 2017

Sales		
Sales revenue		$480,000
Less: Sales returns and allowances	$12,000	
Sales discounts	8,000	20,000
Net sales		460,000
Cost of goods sold		316,000
Gross profit		144,000
Operating expenses		
Salaries and wages expense	64,000	
Utilities expense	17,000	
Advertising expense	16,000	
Depreciation expense	8,000	
Freight-out	7,000	
Insurance expense	2,000	
Total operating expenses		114,000

Other Revenues and Gains

Interest revenue from notes receivable and marketable securities.
Dividend revenue from investments in capital stock.
Rent revenue from subleasing a portion of the store.
Gain from the sale of property, plant, and equipment.

Other Expenses and Losses

Interest expense on notes and loans payable.
Casualty losses from recurring causes, such as vandalism and accidents.
Loss from the sale or abandonment of property, plant, and equipment.
Loss from strikes by employees and suppliers.

Illustration 5-14

LO 5

Multiple-Step

Key Items:

- Net sales
- Gross profit
- Operating expenses
- Nonoperating activities
- **Net income**

PW AUDIO SUPPLY
Income Statement
For the Year Ended December 31, 2017

Sales		
Sales revenue		$480,000
Less: Sales returns and allowances	$12,000	
Sales discounts	8,000	20,000
Net sales		460,000
Cost of goods sold		316,000
Gross profit		144,000
Operating expenses		
Salaries and wages expense	64,000	
Utilities expense	17,000	
Advertising expense	16,000	
Depreciation expense	8,000	
Freight-out	7,000	
Insurance expense	2,000	
Total operating expenses		114,000
Income from operations		30,000
Other revenues and gains		
Interest revenue	3,000	
Gain on disposal of plant assets	600	3,600
Other expenses and losses		
Interest expense	1,800	
Casualty loss from vandalism	200	2,000
Net income		$ 31,600

Illustration 5-14

LO 5

Single-Step Income Statement

- Subtract total expenses from total revenues
- Two reasons for using the single-step format:
  1. Company does not realize any profit until total revenues exceed total expenses.
  2. Format is simpler and easier to read.

Single-Step Income Statement

Illustration 5-15

PW AUDIO SUPPLY
Income Statement
For the Year Ended December 31, 2017

Revenues		
Net sales		$460,000
Interest revenue		3,000
Gain on disposal of plant assets		600
Total revenues		463,600
Expenses		
Cost of goods sold	$316,000	
Operating expenses	114,000	
Interest expense	1,800	
Casualty loss from vandalism	200	
Total expenses		432,000
Net income		$ 31,600

LO 5

Classified Balance Sheet

PW AUDIO SUPPLY
Balance Sheet (Partial)
December 31, 2017

Illustration 5-16

Assets		
Current assets		
Cash		$ 9,500
Accounts receivable		16,100
Inventory		40,000
Prepaid insurance		1,800
Total current assets		67,400
Property, plant, and equipment		
Equipment	$80,000	
Less: Accumulated depreciation—equipment	24,000	56,000
Total assets		$123,400